

Chip-Based Capillary Zone Electrophoresis Mass Spectrometry for Rapid Resolution and Quantitation of Critical Quality Attributes in Protein Biotherapeutics

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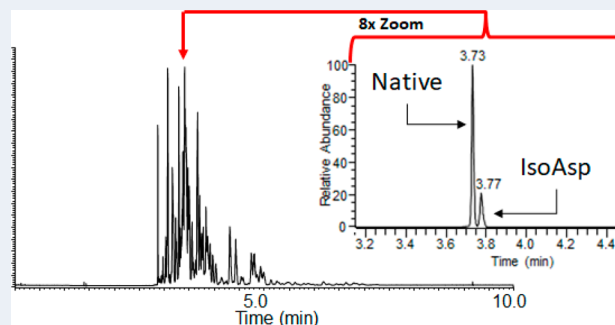


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ABSTRACT: The aspiration of the multi-attribute method (MAM) is to utilize a single mass spectrometry-based method that can measure multiple attributes simultaneously, thus enabling data-driven decisions more quickly and efficiently. However, challenges associated with identifying and quantitating critical quality attributes such as asparagine deamidation and isoaspartic acid using conventional ultrahigh-pressure liquid chromatography (UHPLC) coupled to mass spectrometry have necessitated long gradients to ensure sufficient separation for quantitation. Microfluidic chip-based capillary zone electrophoresis mass spectrometry (CZE-MS) shows potential to enable rapid charge-based separation of peptide mixtures, and this approach was evaluated using multipolypeptide mixtures of synthetic peptides as well as digested protein therapeutics. In these experiments, repeatability, linearity, and peak-to-peak resolution of several peptide families containing asparagine deamidation and/or isoaspartic acid were demonstrated. In addition, a comparison of peptide map results acquired with both UHPLC-MS and CZE-MS for two enzymatically digested biological therapeutics showed comparable sequence coverage and quantitation results between the two approaches. As MAM becomes increasingly utilized for analysis of biological therapeutics, MS instrument demand will rapidly increase, resulting in a bottleneck. A CZE-based separation shows potential to alleviate this bottleneck by drastically increasing MAM throughput while providing results comparable to those acquired using conventional UHPLC separations.



INTRODUCTION

The multi-attribute method (MAM) is a mass-spectrometry-based method of characterizing and quantitating chemical liabilities of biological therapeutics that is consistent with Quality by Design (QbD) principles.^{1–3} While conventional purity assays such as capillary electrophoresis (CE) and cation exchange chromatography (CEX) typically report percentages of nondescript species (e.g., basic species, acidic species, low molecular weight species), MAM can be used to specifically monitor only those quality attributes that may directly affect product quality (e.g., potency and/or safety). In addition, MAM can also be used in lieu of other conventional release methods such as hydrophilic interaction chromatography (HILIC) and enzyme-linked immunosorbent assay (ELISA) to monitor attributes such as glycosylation and molecule identity.⁴

During MAM analysis of biological therapeutics, many peptides containing chemical liabilities, such as oxidation and glycosylation, can be readily resolved from the corresponding unmodified peptide by the mass shift induced by the attribute of interest.⁵ However, the isotopic envelope of a peptide containing isoaspartic acid (isoAsp) or asparagine deamidation overlaps with that of a corresponding native peptide, making

quantitation challenging. Due to the challenges in resolving protein and peptide isomers containing α - and β -linked amino acids, it has been difficult to fully understand the impact of these isomeric species on biological activity. Isoaspartic acid is a naturally occurring β -amino acid that is a result of the degradation of aspartic acid (Asp) and asparagine (Asn). Once isolated from host cells, protein pharmaceuticals are devoid of repair mechanisms and are thus prone to the accumulation of isoAsp upon long-term storage.^{6,7} This degradation product is often considered a critical sequence variant that must be monitored in the control of biotherapeutics to ensure safety and efficacy for patients over the long-term storage of the therapeutic. Therefore, some form of separation of isoAsp and Asp-containing peptides is required to accurately identify and quantitate these attributes.⁸

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Despite the critical roles of Asp and IsoAsp amino acid residues, their structural elucidation and separation have been analytically challenging, limiting a full understanding of the impact of these critical sequence variants to their biological activity. It is further complicated by the low abundance of these isomers. In the biopharmaceutical industry, reversed-phase liquid chromatography (RPLC) combined with mass spectrometry is the most commonly employed separation method.⁹ However, optimizations and analysis using these separations can often be long and tedious. Furthermore, peak assignment is often dictated by elution order since Asp and isoAsp are isobaric. Due to the similar physical properties of these species, differentiation by elution order for identification still proves to be challenging. In an effort to eliminate these slow and tedious steps and provide higher specificity for identification and quantitation, there has been significant interest in developing fragmentation methods that enable chiral discrimination of these peptides by mass spectrometry (MS).^{10,11}

Ultrahigh-pressure liquid chromatography (UHPLC)-based MAM utilizes long gradients to ensure sufficient chromatographic resolution of peptides containing difficult-to-separate modifications. Both deamidation and formation of isoAsp induce charge shifts,¹² and while UHPLC column vendors have made strides in developing reversed-phase columns that can leverage peptide charge in addition to hydrophobicity,^{13,14} capillary electrophoresis-mass spectrometry^{15,16} may offer a more efficient technique for separating charge variants. Coupling capillary electrophoresis with mass spectrometry for proteomics studies has been widely reported in the literature,^{17–20} and this hyphenated approach has found many applications in the characterization of therapeutic proteins and related variants and impurities.^{21–37} Recent reports of rapid analysis afforded of biological molecules using a microfluidic capillary zone electrophoresis unit interfaced with high resolution mass spectrometry (CZE-MS)^{38,39} are particularly attractive for MAM. Rapid analysis time is primarily facilitated by charge-based separation mechanism of CZE, but because no stationary phase is used in CZE-MS analysis, carryover is limited, and column washing and equilibration are not necessary between injections. These time-consuming desalting, washing, and equilibrating steps for UHPLC-based methods can consume up to an hour of instrument time per injection.

In this work, chip-based capillary electrophoresis mass spectrometry (CZE-MS) was used as an integrated microfluidic platform to demonstrate rapid (<10 min) resolution of isoAsp from Asn- and Asp-containing peptides as well as characterization of peptide map samples. The applicability of CZE-MS for MAM analysis for multiple protein systems was also interrogated via the applicability of the system to resolve Asn-, Asp-, and isoAsp-containing peptides over a range of peptide sizes and hydrophobicities. Furthermore, low level nanomolar quantitation for the peptides was demonstrated when resolution was achieved. Two biological therapeutics, an IgG1 monoclonal antibody (mAb) and a bispecific single-chain protein (BiS), were subjected to tryptic digestion, and MAM analyses were compared using conventional UHPLC-MS and CZE-MS. In addition to sequence coverage, 15 critical quality attributes were monitored between the two molecules to evaluate feasibility of implementing CZE-MS for early phase molecule development.

■ EXPERIMENTAL SECTION

Reagents and Materials. All reagents were of the highest purity. Tris–HCl was obtained from Teknova (Hollister, CA), guanidine–HCl, dithiothreitol (DTT), LCMS-grade water, LCMS-grade acetonitrile, trifluoroacetic acid, and formic acid were obtained from ThermoFisher Scientific (Waltham, MA), and sodium iodoacetate (NaIAA) was obtained from Sigma-Aldrich (St. Louis, MO). The biological protein therapeutics analyzed included an IgG1 mAb and BiS that were produced by Amgen, Inc. All synthetic peptides were obtained from Anaspec, Inc., and stock solutions were prepared at 100 μ M concentration in the Peptide Diluent supplied from 908 Devices that contained water, acetonitrile, formic acid, and ammonium acetate. Several solutions were prepared from the stock in concentrations ranging from 10 nM to 30 μ M in the same sample diluent. Equimolar mixtures of peptides were also prepared to assess the resolution that could be obtained in the experimental setup.

Forced Degradation. Thermal degradation was induced by incubating samples in formulation buffer at 40 °C for 4 weeks. Physiological pH degradation was induced by adjusting samples in formulation to pH 7.4 with phosphate-buffered saline (PBS) followed by incubation at 37 °C for 4 weeks. Photodegradation was induced by exposing samples in formulation buffer to 1.2 million lux·h (white light + UV) using a Caron 6445-2 (Marietta, OH).

Enzymatic Digestion. Biological therapeutics were denatured with guanidine–HCl, reduced with DTT, alkylated with NaIAA, and enzymatically digested using methods adapted from Ren et al.⁴⁰ and Wiśniewski et al.⁴¹ For UHPLC-MS analysis, the final peptide concentration was $\sim$ 0.2 mg/mL. Because the CZE-MS system was not compatible with tris buffer, samples analyzed with CZE-MS were buffer exchanged into 80% acetonitrile, 0.1% trifluoroacetic acid using C18 MicroSpin columns (The Nest Group, Southborough, MA), resulting in a final peptide concentration of $\sim$ 0.1 mg/mL.

UHPLC-MS Analysis of Digested Protein Samples. Digested samples were analyzed using a Thermo Scientific (Waltham, MA) U-3000 UPLC system connected in-line to a Thermo Scientific Q-Exactive. A reversed-phase HPLC column (Zorbax SB-C18 Rapid Resolution HD, 2.1 $\times$ 150 mm, 1.8 μ m; Agilent, Santa Clara, CA) was used to separate the peptides with the column temperature at 50 °C using mobile phase A: 0.1% formic acid (v/v) in water and mobile phase B: 0.1% formic acid in acetonitrile. The gradient (hold %B at 1.0% for 10 min, then 1–10% B in 1 min and 10–36%B in 146 min) was performed at 0.25 mL/min. Approximately 3 μ g of each sample was injected for each analysis. The mass spectrometer was operated in positive mode using data-dependent acquisition (top 6). Full MS resolution was set to 140000 at m/z 200 with a scan range of 200–2000 m/z . Full MS automatic gain control was set to 1E6, and maximum IT was set to 120 ms. MS2 data were acquired at 17,500 resolution using stepped normalized collision energy of 20, 24, and 28 with automatic gain control set to 1E5 and dynamic exclusion of 10.0 s. Peptides with unassigned charge states or charges greater than 8 were not selected for MS2 analysis. Data was acquired for a single injection for each sample, though reproducibility was assessed internally during method development and was consistent with previous reports.^{42,43}

CZE-MS Analysis of Digested Protein Samples and Synthetic Peptides. Digested samples and synthetic peptides

Table 1. Overview of Synthetic Peptides Evaluated with CZE-MS

Peptide	# Resi- dues	Native MW (Da)	Native GRAVY	Migration Time (min)	Peak Width (min)	Reso- lution	# of Peptides Per Family	# Observed Peaks
XXXXXXXXDX	10	916	0.21	3.70	0.04	-	2	2
XXXXXXXXD[iso]X				3.80	0.04	2.50		
XXXXXDXXXXXXXXXX	14	1677	-0.46	3.41	0.04	-	2	2
XXXXXD[iso]XXXXXXXXXX				3.53	0.04	3.00		
XXXXXDXXXXXXXXXXXXXX	21	2329	-1.23	3.81	0.04	-	2	2
XXXXXD[iso]XXXXXXXXXXXXXX				3.92	0.04	2.75		
XXXXXD[iso]XXXXXXXXXX	17	1873	-0.13	4.45	0.04	-	3	3
XXXXXD[iso]XDXXXXXXXXXX				4.52	0.04	1.75		
XXXXXD[iso]XXXXXXXXXX				4.58	0.04	1.50		
XXXXNXXXX	9	1084	-1.06	3.72	0.04	-	3	3
XXXXDXXXX				3.76	0.04	1.00		
XXXXD[iso]XXXX				3.84	0.04	2.00		
XXNXXNXXX	9	1007	-0.64	3.73	0.04	-	5	3
XXNXXDXXX				3.76	0.04	0.75		
XXNXXD[iso]XXX				3.76				
XXDXXNXXX				3.86	0.04	2.50		
XXD[iso]XXNXXX				3.86				
XXXXXXXXNXXXXXXXXXXXXX	27	3070	-1.25	3.49	0.07	-	7	2
XXXXXXXXDNNXXXXXXXXXXXXX				3.53				
XXXXXXXXD[iso]NNXXXXXXXXXXXXX				3.52				
XXXXXXXXNDNXXXXXXXXXXXXX				3.58	0.07	1.10		
XXXXXXXXND[iso]NXXXXXXXXXXXXX				3.61				
XXXXXXXXNNDXXXXXXXXXXXXX				3.61				
XXXXXXXXNND[iso]XXXXXXXXXXXXX				3.57				
XXXXXXXXXXXXNXXNXX	22	2543	-1.26	5.10	0.10	-	7	2
XXXXXXXXXXXXDXXNXX				5.08				
XXXXXXXXXXXXD[iso]XXNXX				5.11				
XXXXXXXXXXXXNXXDNXX				5.15	0.09	1.10		
XXXXXXXXXXXXNXXD[iso]NXX				5.31				
XXXXXXXXXXXXNXXNDXX				5.29				
XXXXXXXXXXXXNXXND[iso]XX				5.16				

were analyzed using a 908 Devices (Boston, MA) ZipChip system connected in-line to a Thermo Scientific Q-Exactive Biopharma in triplicate. Peptides were separated on HR chips using the “Peptide Background Electrolyte” solution obtained from 908 Devices that contained water, acetonitrile, and formic acid. A unique HR chip was dedicated to BiS and mAb samples, respectively. For all samples, the ZipChip was operated using vendor recommended conditions without further optimization; field strength was 500 V/cm, load time was 30 s, and pressure assist was enabled. For the synthetic peptides, a total run time of 5 min was used, and full MS spectra were collected from m/z 100 to 1500 in positive-ion mode. For the digested protein samples, a total run time of 10

min was used, and the mass spectrometer was operated in positive mode with a scan range of m/z 300–2000. Data were acquired in both data-dependent mode (top 3, resolution = 35000 at m/z 200) for peptide identification and sequence coverage determination and MS-only mode (resolution = 17000 at m/z 200) for relative quantitation. For Full MS scans, automatic gain control was set to 1E6, and maximum IT was set to 50 ms. For data-dependent analysis, MS2 were acquired at 17500 resolution using stepped normalized collision energy of 20, 24, and 28 with automatic gain control set to 1E5 and dynamic exclusion of 0.9 s. Peptides with unassigned charge states or charges greater than 8 were not selected for MS2 analysis. Limited carryover was observed between injections, so

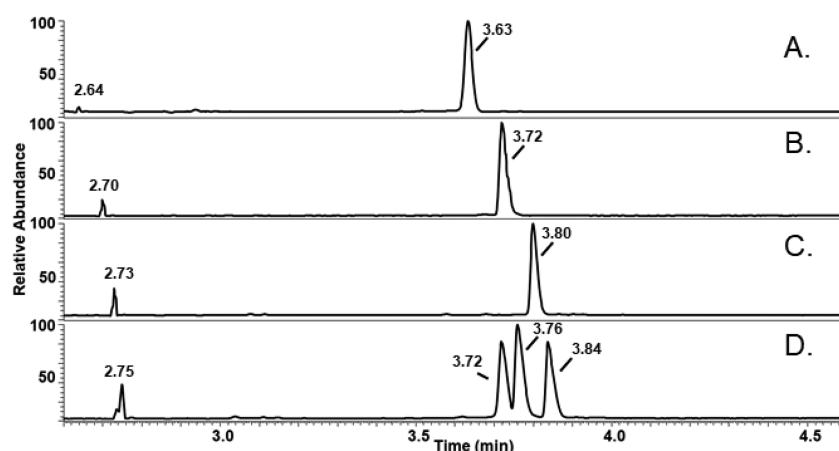


Figure 1. Total ion electropherograms of synthetic peptides (A) XXXXNXXXX, (B) XXXXDXXXX, (C) XXXXD[iso]XXXX, and (D) and equimolar mixture of all three peptides.

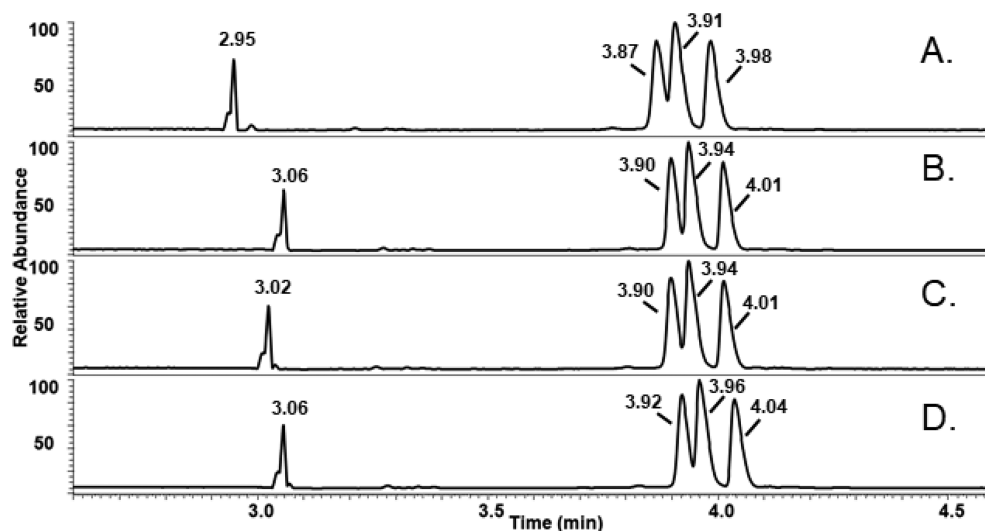


Figure 2. Total ion electropherograms of four replicate injections of an equimolar mixture of synthetic peptides XXXXNXXXX, XXXXDXXXX, XXXXD[iso]XXXX.

wash injections were not performed after sample injections (Supporting Information).

Data Analysis. Peptide identification, sequence coverage analysis, and site-specific post-translation modification identification was performed using MassAnalyzer v. 3.02⁴⁴ specifying 15 ppm mass accuracy, a S/N threshold of 20, and minimum confidence of 0.95. Only precursors with corresponding MS2 were searched. Relative quantitation was performed using Skyline v. 3.7.0.11317.⁴⁵ For each molecule, a workbook was created containing native and modified versions of all peptides identified with quality attributes of interest. Total area of peptides was determined by extracting ion chromatograms of two charge states per peptide (three isotopes per charge state). For the digested samples, modification percentages were calculated by dividing the total area of the modified peptide by the sum of total areas from the modified and native peptides. New peak detection was performed on CZE-MS data acquired for digested peptides with the Non-Targeted MS Processing component of Chromeleon v7.2.10 (Thermo Scientific, Waltham, MA) with the following parameters: m/z width = 15.00 ppm, retention time start = 2.00 min, retention time end = 7.00 min, frame width = 0.5 min, maximum number of frames = 10000, peak intensity threshold =

autocompute TIC 1.00%. Frame filters were specified to match all of the following rules: PR element = 0, PR size > 1, charge ≥ 1 , charge ≤ 5 , and ratio is not between 0.10000 and 10.00000. For the analysis of synthetic peptides, the area for each peptide was used to build a calibration curve which informed the limit of quantitation (LOQ) of the method. The LOQ was determined by assessing where the S/N of the peptide was greater than 10.

RESULTS AND DISCUSSION

Applicability to Multiple Peptide Systems. Initial assessment of the CZE-MS system was performed using a set of synthetic peptides corresponding to tryptic peptides previously observed in biological therapeutics evaluated internally (Table 1). These peptides all contained at least one Asn or Asp residue that was observed to be susceptible to deamidation or isomerization, respectfully, following forced degradation studies of the corresponding biological therapeutics. Native and modified versions of each peptide were obtained; because Asn deamidation can lead to the formation of both Asp and isoAsp acid through a succinimide intermediate,^{46,47} both variants were obtained for each peptide

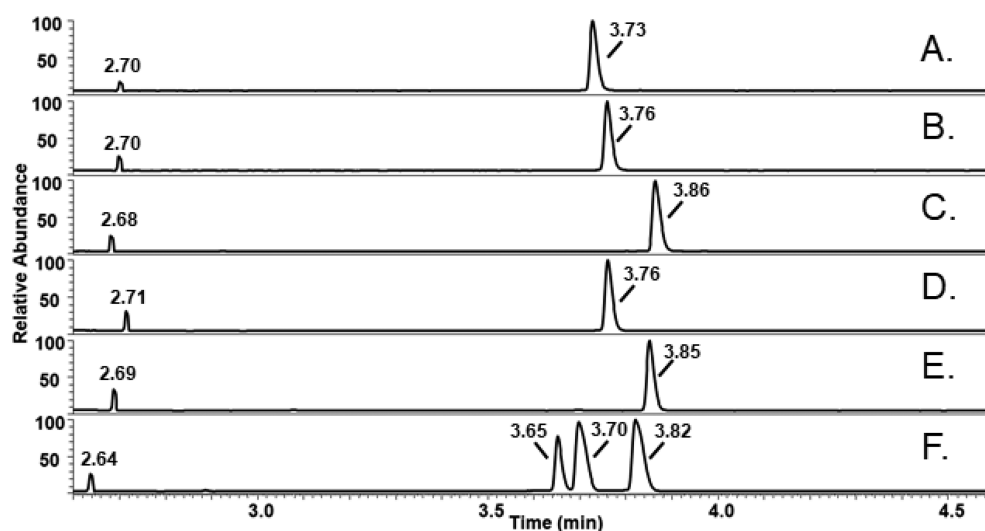


Figure 3. Total ion electropherograms of synthetic peptides (A) XXNXXNXXX, (B) XXNXXDXXX, (C) XXNXXD[iso]XXX, (D) XXDXXNXXX, (E) XXD[iso]XXNXXX, and (F) an equimolar mixture of all five peptides.

containing Asn residues. Some of these peptides contained multiple Asn or Asp residues. While internal investigations of the corresponding biological therapeutic did not always show evidence of all potential variants in peptides containing multiple asparagine or aspartic acid residues, synthetic peptides were still obtained in order to push the separation limits of the CZE. In total, 31 synthetic peptides corresponding to 8 peptide families were evaluated. These peptides encompassed a range of physical properties typically observed during peptide map experiments. The peptides ranged from 9 to 27 residues, the molecular weight of the native peptide ranged from 916 to 3070 Da, and the hydrophobicity as determined by GRAVY (grand average of hydrophobicity)⁴⁸ score ranged from -1.26 (hydrophilic) to 0.21 (slightly hydrophobic).

Synthetic peptides were injected individually and as an equimolar mixture of all variants corresponding to a single peptide family as illustrated in Figure 1, which highlights a representative peptide family (native peptide XXXXNXXXX). In Figure 1, all three peptide variants are clearly resolved in the equimolar mixture, and the charge variants of the peptide migrated in the order of Asn $\rightarrow$ Asp $\rightarrow$ isoAsp. This was consistent with the expected charge-based separation (basic $\rightarrow$ acidic); previous studies evaluating physical properties of Asn, Asp, and isoAsp and charge-based separation of these species report that the migration order is due to the negative charge induced by deamidation and slight differences in the pKas of Asp and isoAsp.^{49–53} This migration order was observed in all peptides evaluated in this study.

Repeatability of this equimolar mixture was evaluated in a separate sequence, and Figure 2 collects four injections for the XXXXNXXXX peptide family. For these injections, migration times for each of the three peaks had precision better than 1.0% RSD. A general trend of increasing migration times with increasing injections was not observed as slight shifts in migration time were irregular. Migration times from these peptides were also evaluated across multiple days, and the precision of the migration time for each of the three peaks was still better than 2.5%, which was consistent with UHPLC-based MAM internal guidelines. Similarly, peak area was also evaluated across the four injections, and for each variant, peak area precision was better than 5.0% RSD. Peak area precision

across multiple days was also better than 10.0% RSD, which again was consistent with UHPLC-based MAM internal guidelines.

Linearity was also evaluated for these peptides (Supporting Information). Equimolar mixtures of each peptide family were prepared at concentrations from 10 nM to 30 μ M, and peak areas were determined for each variant and each concentration. For each of the variants, an R^2 greater than 0.98 was observed across a concentration range of ~ 2 orders of magnitude, and based on the slope of the corresponding linearity curves, an LOQ at ~ 10 nM for all three peptides could readily be achieved. These results suggest that with a proper calibration curve, this methodology is suitable for quantitation to low nanomolar levels. This is particularly applicable to studies involving peptide therapeutics where reporting a weight percentage of an impurity may be more appropriate than the relative quantitation approach typically utilized for MAM.

Of the eight peptide families evaluated, five of these had three or fewer variants. For each variant corresponding to these peptide families, the peak width was 0.04 min, peak-to-peak resolution greater than 1.0 was observed, and the number of peaks observed was equal to the number of peptides within a particular peptide family (Table 1). This suggested that for peptides exhibiting three or fewer charge variants (either a single asparagine residue or up to two aspartic acid residues), this CZE-MS approach could be used for site-specific characterization of these degradants. Resolution became more challenging with peptide families containing more than three charge variants with either no observed resolution or resolution less than 1.0. Equimolar mixtures of each of the three peptide families with more than three variants resulted in significantly fewer observed peaks compared to the total number of peptides per family. For example, native peptide XXNXXNXXX has two potential Asn deamidation sites, and a total of five peptides were evaluated for this peptide family. While the signal for each of the five variants was comparable when injected individually, only three peaks were observed when the equimolar mixture of all five variants was injected (Figure 3). Additionally, peak areas of the two later migrating peaks (migration times of 3.70 and 3.82 min) were roughly double the peak area of the peak migrating at 3.65 min. This

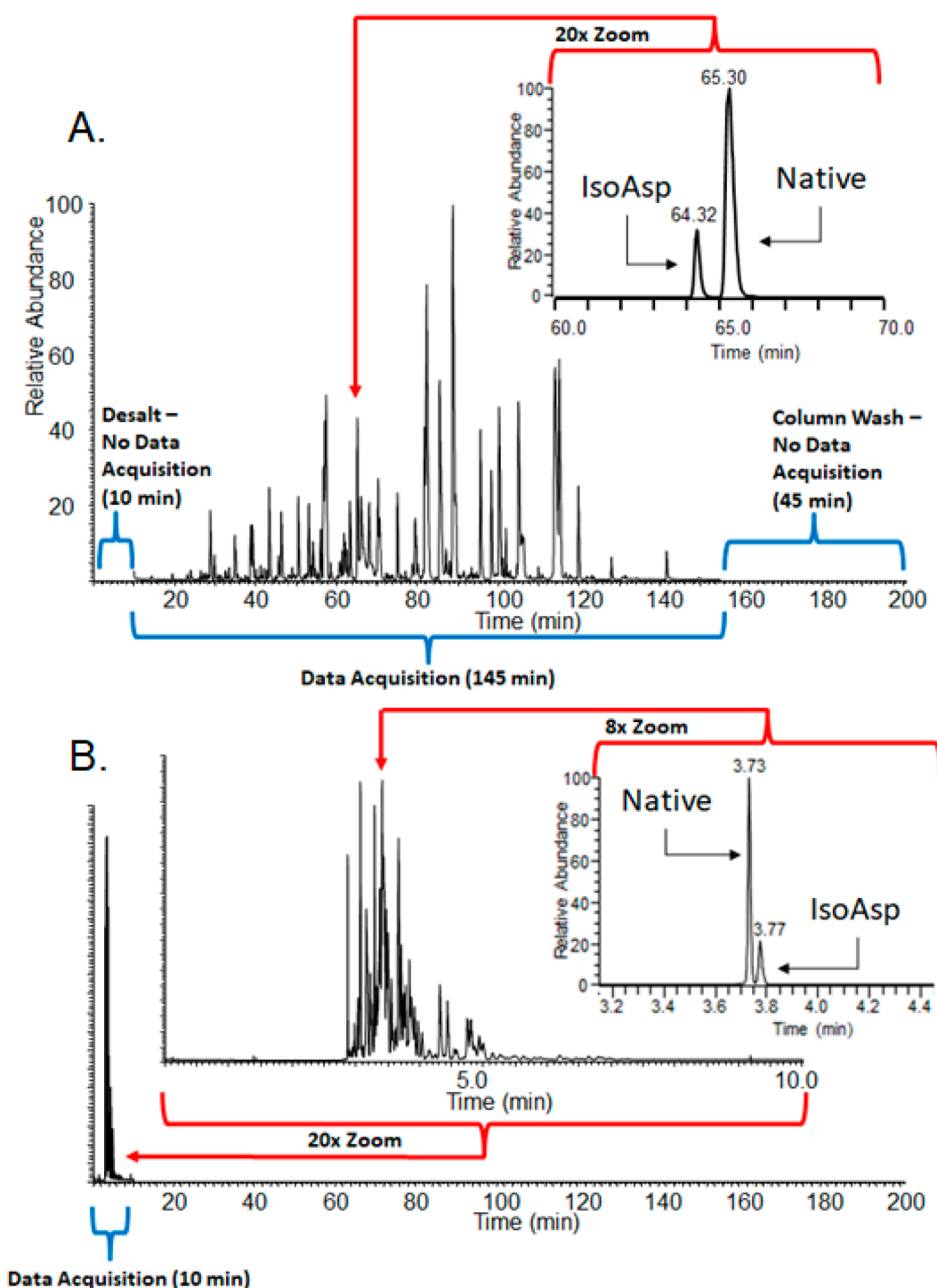


Figure 4. Total ion chromatogram of peptide map of BiS with (A) conventional UHPLC-MS method and (B) total ion electropherogram of the same sample using CZE-MS. The zoomed insets correspond to the extracted ion chromatogram (A) and extracted ion electropherogram (B) for the native and modified versions of the peptide of interest.

observation is consistent with comigration of the peptides containing Asp (XXDXXXNXXX, XXNXXDXXX) at 3.70 min and peptides containing isoAsp (XXD[iso]XXNXXX, XXNX-XD[iso]XXX) at 3.82 min in the equimolar mixture injection. These results demonstrate that only three variants could be differentiated in a mixture of 5 peptides as peptides modified at different sites could not be distinguished.

Of the peptide families evaluated (Table 1), it should be noted that peptide XXNXXXNXXX is relatively small, consisting

of nine residues (molecular weight of 1007 Da). The two remaining families containing more than three variants were significantly larger (≥ 22 residues, 2500 Da), and Asn residues were all in close proximity to each other; for these peptide families, only two peaks were observed, and site-specific characterization and quantitation of each variant in the mixture was not achieved. However, results from these experiments could still be used to evaluate regions of the molecule that may be susceptible to degradation and potentially inform decisions

about molecule stability even if the exact site of the degradation could not be determined in these more challenging peptide families. Additionally, these data were collected using vendor recommended conditions; better resolution may potentially be achieved with further optimization of CZE instrument parameters.

Peptide Identification Using UHPLC-MS vs CZE-MS in Protein Digest. Results of peptide identification analysis of a tryptic digest of unstressed IgG1 mAb and BiS samples on a conventional UHPLC-MS method were compared to results obtained with the CZE-MS system. Figure 4 shows the comparison of total ion chromatogram and total ion electropherogram for the mAb digest. Strikingly, the data acquisition time was cut from 145 min to only 10 min with the CZE-MS methodology. Despite the significant decrease in acquisition time when using the CZE-MS system, sequence coverage for both molecules were slightly higher for CZE-MS samples compared to UHPLC-MS analysis for both digests (Table 2). Inspection of sequence coverage maps showed that

not being retained by the column in these experiments. This is consistent with a previous comparison of peptide map sequence coverage acquired using LC-MS and CE-MS.²²

To compensate for the significant reduction in MS analysis time in the CZE-MS experiment, it was critical to optimize MS parameters to enable accurate peptide identification in this rapid analysis system to maximize sequence coverage. Since peak widths are relatively broad with UHPLC-MS analysis (15–30 s), it is possible to acquire higher resolution full MS scans and data-dependent MS2 scans while maintaining a sufficient number of full MS scans across the peak for quantitation (ideally >10 scans per peak). For example, using UHPLC-MS, one representative modified peptide (mAb Met oxidation 1) has a peak width of 19.2 s, resulting in 35 full MS scans performed across this peak at 140,000 resolution plus up to 6 data dependent MS2 scans per full MS scan. Extremely narrow peaks observed with CZE-MS necessitate significantly higher duty cycles. Using CZE-MS, this same peptide was observed with a peak width of 2.4 s allowing for only six full MS scans performed across the peak at 35000 resolution plus up to three data-dependent MS2 scans per full MS scan. CZE-MS data-dependent analysis was initially performed using resolution of 17,500 to increase duty cycle, but at this resolution, the charge state of a 6 kDa heavy chain peptide conserved among this class of proteins was consistently not identified. As a result, the peptide was not selected for MS2 fragmentation, and sequence coverage for this molecule at resolution of 17500 was significantly lower than that observed with UHPLC-MS. Therefore, it was critical that a resolution of at least 35000 was used for peptide identification for the

Table 2. Sequence Coverages Observed for Peptide Mapping Experiments of Biological Therapeutics

Separation	mAb Chain 1 (%)	mAb Chain 2 (%)	BiS (%)
UHPLC-MS	93.3	93.3	93.6
CZE-MS	98.0	97.2	98.0

these differences were primarily due to small peptides (three to five amino acid residues) that were not identified during UHPLC-MS analysis, likely a result of these smaller peptides

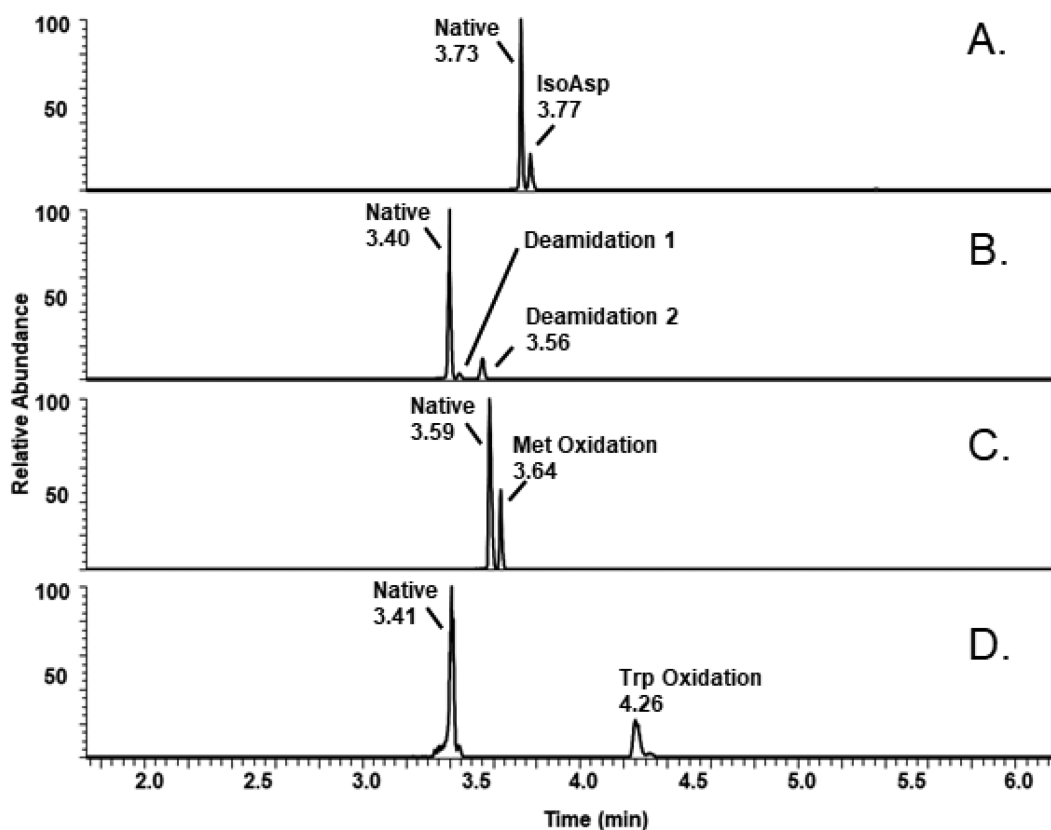


Figure 5. Extracted ion chromatograms of (A) native and isoAsp peptide, (B), native and deamidation peptides, (C) native and methionine oxidation peptides, and (D) native and tryptophan oxidation peptides. Annotated MSMS for modified peptides in this figure are collected in the Supporting Information.

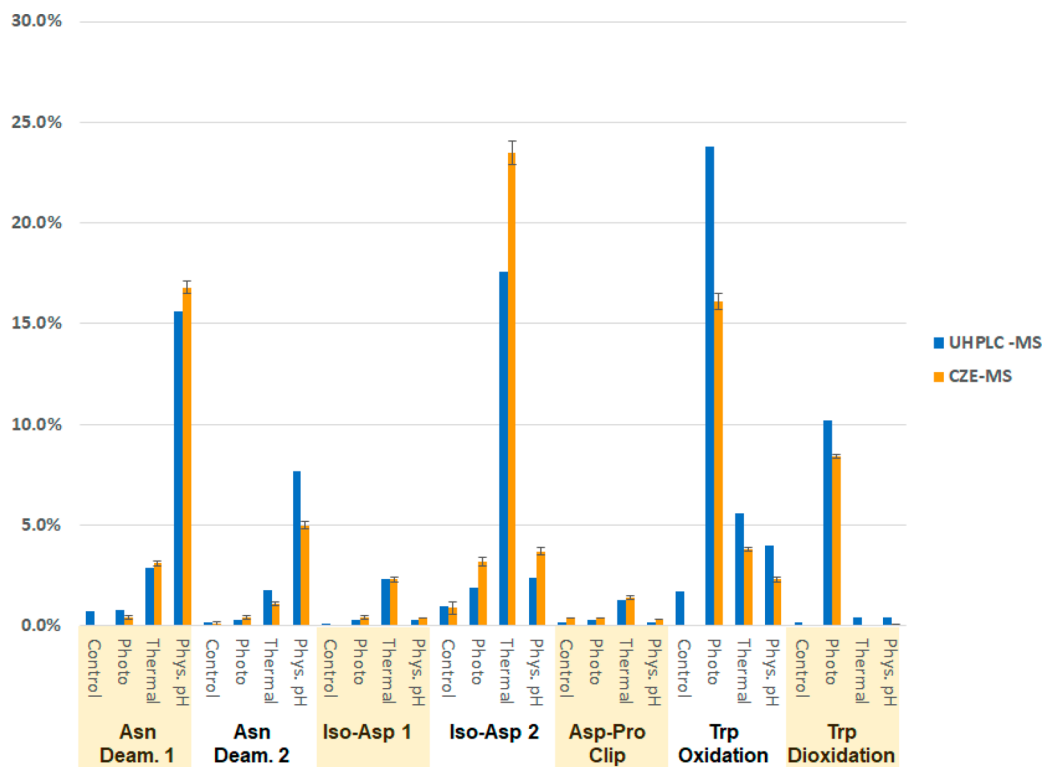


Figure 6. Comparison of attribute quantitation using UHPLC-MS and CZE-MS for BiS.

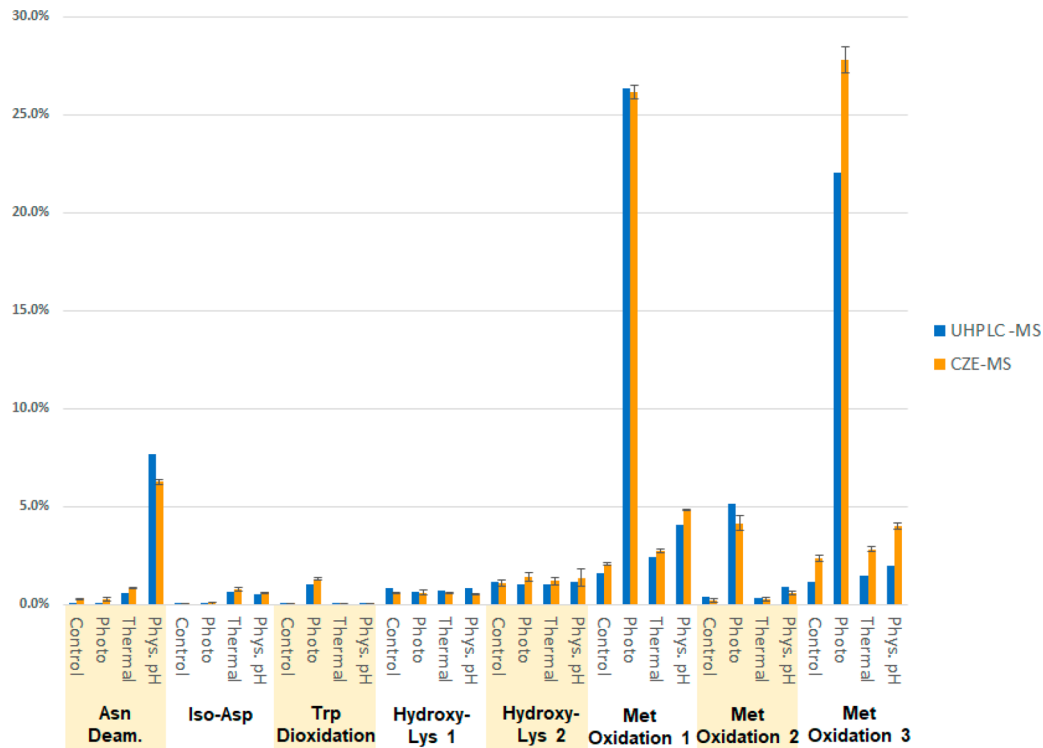


Figure 7. Comparison of attribute quantitation using UHPLC-MS and CZE-MS for mAb.

comparison performed. While these MS parameters were optimized for identification purposes for the CZE-MS experiment, a MS-only experiment similar to conventional MAM was performed to enable a sufficient number of full MS scans across the peak for quantitation purposes for MAM analysis. At 17500 resolution with no data-dependent scans, 28

full MS scans were performed across the same peak with a width of 2.4 s, which is more consistent with the number of scans per peak observed with UHPLC-MS. A characterization-focused experiment may utilize two sequential injections (data-dependent acquisition with MS1 resolution of 35,000 followed MS1-only acquisition with MS1 resolution of 17500) to both

identify and quantitate peptides; even with two 10 min injections per sample, throughput would still be considerably higher than conventional UHPLC-MS analysis. Interfacing rapid CZE separations with the Bruker timsTOF Pro,⁵⁴ with reported scans speeds of up to 100 Hz, may also allow for sufficient MS1 resolution and points across the peak to combine peptide identification and quantitation in a single injection.

Samples were subjected to thermal stress, physiological pH stress, and photostress, and corresponding digests were also analyzed to identify the chemical liabilities of each protein. Figure 5 shows representative extracted ion chromatograms (EIC) for several types of attributes known to impact either molecule potency and/or safety that are typically monitored with MAM, including aspartic acid isomerization, asparagine deamidation, methionine oxidation, and tryptophan oxidation. Peptide-level separation of additional modifications relevant to biological therapeutics using this instrumentation has also been reported.⁵⁵ Of note is that peptides containing isoAsp and Asn deamidation were separated from corresponding native peptides with baseline or near-baseline resolution using a 10 min CZE-MS method. While these attributes are typically the most challenging for UHPLC-based MAM, CZE-MS was also capable of rapidly separating other attributes known to impact molecule quality, including both methionine and tryptophan oxidation. Furthermore, EICs for BiS isoAsp 2 in a control and thermally degraded sample demonstrate that observed modification levels were not artifacts of the separation and that observed differences in isoaspartic acid formation were a direct result of the degradation conditions.

Relative Quantitation of Chemical Liabilities in Peptide Map. Similar trends in susceptibility to degradation were observed regardless of how peptides were separated (Figures 6 and 7). For example, BiS isoAsp 1 was observed in the control sample at 0.1% and 0.0% when separated using CZE-MS and UHPLC, respectively. After thermal degradation, isomerization levels of 2.3% (CZE-MS) and 2.3% (UHPLC) were observed for this residue. Similarly, BiS Asn deamidation 1 was observed in the control sample at 0.0% (CZE-MS) and 0.7% (UHPLC). After degradation at physiological pH, deamidation levels of 16.8% (CZE-MS) and 15.6% (UHPLC) were observed for this residue. While most attributes were observed at similar levels, some significant differences were observed. BiS isoAsp 2 was observed at levels ~6% higher in CZE-MS experiments relative to UHPLC-MS experiments, and BiS Trp oxidation was observed at ~8% higher in UHPLC-MS experiments relative to CZE-MS experiments (Figure 6). On the basis of these results, it was unclear whether these results were due to the amount of time that passed between UHPLC-MS and CZE-MS analysis, different mass spectrometers used for analysis (Q Exactive vs Q Exactive Biopharma), elevated column temperature and long UHPLC run times, or fundamental differences in separation mechanisms.

New Peak Detection Analysis. New peak detection is a critical aspect of MAM as it allows analysts to quickly detect and identify variants that may directly affect product quality, even if these variants are not specifically monitored as part of the method. Triplet injections described earlier in this manuscript were injected sequentially in the sequence. For example, all three digested BiS control injections were completed before starting the three digested BiS thermal stress samples. New peak detection analysis was performed on

the three control injections for mAb and BiS samples; injections 2 and 3 were evaluated relative to injection 1, and injection 1 was evaluated relative to injection 3. New peak detection for all stressed samples was performed relative to control injection 1. Using a frame time width of 0.5 min (± 0.25 min relative to reference) in the Non-Targeted MS Processing software, all control injections passed new peak detection. All stressed BiS samples failed new peak detection, which was expected due to increase of chemical liabilities resulting from the stressed conditions. However, while mAb photostress samples failed new peak detection, thermal stress and physiological pH stress samples passed new peak detection. This was unexpected because on the basis of Figure 7, Asn deamidation clearly exceeded the ratio threshold specified in the Non-Targeted MS processing parameters (5-fold change in peak area relative to reference). Inspection of components triggering the failures showed that these were primarily oxidized peptides. BiS stressed samples and mAb photostress samples all had at least one oxidized species that increased beyond the Ratio threshold, but oxidized species in mAb thermal stress and physiological stress did not increase to this degree. Oxidation induces a + 16 Da mass shift to the peptide relative to the native peptide, and this was readily flagged as outside of the m/z width window (ppm). IsoAsp is isobaric with the native peptide, and the ^{12}C isotope of a deamidated peptide often has a predicted m/z similar to the ^{13}C isotope of the corresponding native peptide; for example, the predicted m/z of the ^{12}C isotope of the deamidated peptide in Figure 7 differs from the predicted m/z of the ^{13}C isotope of the corresponding native peptide by 7.5 ppm. As such, increased levels of these modified peptides did not result in failure of the new peak detection. In UHPLC-based MAM, the retention time shift induced by these attributes is typically outside of the frame time width parameter, but due to rapid separations and variations in migration times, decreasing the frame time width time beyond 0.5 min resulted in failure of new peak detection for the control samples. In this case, the primary strength of CZE-MS was a limitation. While this approach may be well-suited to molecule assessment and characterization experiments, shifting migration times will need to be addressed prior to implementation of this approach in a QC environment.

Attribute Quantitation Repeatability. Each standard HR chip used in this evaluation was vendor limited to 125 injections. In addition to the triplet injections described earlier in this manuscript, a fourth set of injections (one injection each of control, thermal stress, physiological pH stress, and photostress) was performed after the chip had nearly reached the maximum number of injections. Seventy-three injections were performed between the first of the triplicate mAb control injections and the control injection corresponding to the fourth set of injections (~10 h), and 97 injections were performed between the first of the triplicate BiS control injections and the control injection corresponding to the fourth set of injections (~14 h). New peak detection analysis was performed on the control injections relative to the first control from the initial triplet injections, and both the mAb and BiS control samples passed. Attributes assessed in Figures 6 and 7 were quantitated in samples in the fourth set of injections, and quantitation results from the first set of injections was plotted relative to the results from the fourth set of injections (Supporting Information). R^2 values of 0.9977 and 0.9886 were observed for mAb and BiS samples,

respectively. While chip-to-chip variation was not evaluated in these experiments, this initial evaluation suggests that quantitation results are repeatable over the lifecycle of the chip.

CONCLUSIONS

As MAM becomes increasingly utilized for analysis of biological therapeutics, MS instrument demand will rapidly increase, resulting in a bottleneck. This is particularly evident during lead candidate selection, where multiple candidates in multiple formulations must be evaluated following exposure to multiple stress conditions. Results presented here indicate that rapid separation and quantitation of wild-type peptides containing asparagine from known degradation products where the asparagine amino acid converts to aspartic acid and isoaspartic acid can be achieved in less than 5 min; a CZE-based separation shows potential to improve efficiency by increasing MAM throughput by up to 20x (10 min vs 200 min) while providing results comparable to those acquired using conventional UHPLC separations. A lead molecule selection campaign may assess up to 60 unique samples; implementing MAM with CZE-MS can reduce data acquisition time from over 8 days to an overnight experiment. Robustness of MAM using CZE-MS is still being evaluated, so while it is unclear at this point whether this technology is appropriate for GxP release testing, CZE-MS using ZipChip instrumentation shows great potential to streamline early phase molecule assessment, directly impacting development timelines by expediting the time necessary to select a lead candidate.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jasms.0c00424>.

Evaluation of carryover, MSMS spectra, comparison of attribute quantitation between Injection Group 1 and Injection Group 4 (PDF)

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Notes

The authors declare no competing financial interest.

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